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Adrenal

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Spontaneous Adrenal Hemorrhage with Partial Adrenal Insufficiency in a Patient Anticoagulated with Apixaban for Antiphospholipid Syndrome

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Background: Addison's disease secondary to adrenal hemorrhage is a rare complication of antiphospholipid syndrome (APLS). Anticoagulation with a direct oral

anticoagulant (DOAC) might increase the risk of adrenal hemorrhage. We report a patient who developed bilateral adrenal hemorrhage after starting apixaban for deep venous thrombosis (DVT) and pulmonary embolism (PE) in the setting of latent APLS.

Clinical Case: A 45-year-old man without significant past medical history presented with a left lower extremity DVT and PE. He had no personal or family history of previous trauma, bleeding or clotting events. He was discharged on apixaban. Three days later he was readmitted with bilateral flank pain. Computerized tomography (CT) imaging showed bilateral adrenal gland edema. He was hemodynamically stable. Due to persistent right flank pain and concern for ileus, CT was repeated and revealed findings concerning for adrenal hemorrhage. He subsequently complained of asthenia. Further testing revealed a reduced morning cortisol of 1.4 ug/dL (reference: 6.0 - 18.4 ug/dL) and an elevated morning ACTH of 149 pg/mL (7.2-63.3 pg/mL). Plasma dehydroepiandrosterone sulphate (DHEA-S) was reduced at 39 mcg/dL (70 - 495 mcg/dL), and epinephrine and norepinephrine were undetectable. Morning plasma aldosterone was undetectable and renin activity was 0.76 ng/mL/h (0.25 - 5.82 ng/mL/h). He was started on stress dose steroids and discharged on maintenance hydrocortisone without reinitiating anticoagulation.

Two weeks later, he re-presented with a right lower extremity DVT. An APLS panel revealed elevated anti-beta2 glycoprotein 1 IgG and anti-cardiolipin IgG. Laboratory work-up for other causes of thrombosis, including inherited and acquired thrombophilias, anatomical vascular obstruction, paroxysmal nocturnal hemoglobinuria (PNH), heparin-induced thrombocytopenia (HIT), and myeloproliferative neoplasms (MPN), was unrevealing. Warfarin was started with an international normalized ratio (INR) target range of 2-3. Hydrocortisone was continued at a total daily dose of 30mg. Since discharge, his morning cortisol levels have been low at 4.4-4.9 ug/dL and ACTH has remained high at 92-137 pg/mL, but aldosterone has been variable at <1 to 9 ng/dL, and renin has been normal at 0.35 ng/mL/h. This suggests what transpired had been microthrombi-associated intra-adrenal hemorrhage involving the zona fasciculata with sparing of the glomerulosa.

Conclusion: This case highlights DOAC-associated adrenal hemorrhage in a patient with underlying APLS, an under-appreciated complication. The extent to which the various layers of the adrenal gland are affected may vary. Since guidelines on the management of patients with partial adrenal insufficiency are lacking, further reporting is needed on long term outcomes and adrenal function of these patients.

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